



CREATIVE TECHNO COLLEGE, ANGUL

PROJECT SYNOPSIS

1. Project Topic: Chatbot Using C++

2. Group members:

- Asutosh Sahu
- Sahil Setha
- Subham Behera – 1
- Subhashree Sahu

3. Course: BCA 2nd Sem

4. Year: 2025-2029

5. Guide Name: Mr. Swastik Ranjan Sahoo

6. Submission Date: 23 September 2026

7. Project Overview:

- Web-based chatbot with C++ backend and HTML, CSS, JavaScript interface.
- Handles greetings, common questions, and FAQs using keyword matching.
- Uses a file-based knowledge base for predefined answers.
- Unanswered queries are processed using the Groq API and LLM.
- Maintains chat history and export in txt as file.
- Supports TXT, DOCX, and image uploads for analysis.
- Includes multimodal AI for image processing.
- Provides Admin Mode for FAQ and history management.
- Demonstrates C++, file handling, API integration, web development, and basic security.
- Provides a simple, interactive, and user-friendly chatbot experience.

8. Objective: Clearly state the goals of the project

- To develop a web-based conversational chatbot using a C++ backend.
- To process user input using keyword matching and a file-based knowledge system.
- To provide AI-based responses through the Groq API with conversation context.
- To maintain conversation history and manage questions and answers using file handling.
- To support TXT/DOCX/image analysis and basic admin controls.

9. Methodology: Provide a step-by-step plan for how the group intends to complete the project

- Design the chatbot interface using HTML, CSS, and JavaScript and connect it to the C++ server.
- Implement keyword rules, predefined FAQs, text normalization, and token-overlap matching.
- Integrate the Groq API for open-ended responses and conversation-context handling.
- Implement TXT/DOCX text extraction, image analysis, and response regeneration.
- Store conversations in chat_history.txt and manage FAQs through knowledge.txt.
- Test the complete system, including chat, file upload, history, knowledge management, and error handling.

10. References:

- C++ Documentation; cpp-http lib Documentation; nlohmann/json Documentation; libcurl Documentation; Groq API Documentation; MDN Web Docs; HTML/CSS/JavaScript Documentation; File Handling and String Processing Resources.

Student Signatures

Sign. Of Guide with date